

Front ends (Frontend trait)

Raw ASL (JSON)
deployed artifact

Typed builder DSL
declared by construction

**CNCF Serverless
Workflow DSL**

**CloudFormation / SAM
/ CDK synth**
deployed artifact

Annotation sources

Sidecar (TOML)
declared $\Rightarrow$ errors

Inference
naming / resource
conventions $\Rightarrow$ warnings

Legend

- green – sound ASL facts
- amber – under-approximating native warnings
- purple – provenance-graded semantic premises
- $\rightarrow$ solid – lowering / data path
- $\cdots$ dashed – reads facts / cross-check

StepCheck IR
format-agnostic state graph
Task · Choice · Parallel · Map
Wait · Pass · Succeed · Fail

control flow + Catch edges
JSONPath I/O fields
retry / catch policies
nested sub-machines
resolved-annotation slot

Annotation layer
resolve: sidecar $\rightarrow$ inference
schema names $\rightarrow$ field sets
idempotency · persistence
compensation · typestate
explicit always wins

lower

declared / inferred facts

Pass manager · one deployed-artifact IR ·
eight provenance-graded passes

NATIVE FACTS · SOUND ASL-only · no false positives

Structural
reachability / well-formedness
SC0001-0012

Data-flow provenance
abstract interpretation
SC1001-03 · SC1101 · SC1110

NATIVE FACTS · CONSERVATIVE AWS-resource / temporal warnings

Concurrency interference
SC5001-5003

Temporal / retry budget
SC6001-6003

SEMANTIC PREMISES declared $\Rightarrow$ error · inferred $\Rightarrow$ warning

Typed contract
SC1010

Typestate
SC2001-2002

Retry safety
SC3001

Compensation / Saga
own-handler + downstream
SC4001-02 · SC4010 · SC4011

verified IR

findings

ASL emitter
verified IR $\Rightarrow$ deployable
ASL JSON

AWS Step Functions
runs unmodified

Diagnostics engine
code · severity · state · fix-hint

severity by provenance:
declared $\Rightarrow$ error
inferred $\Rightarrow$ warning

terminal or JSON output

cross-check SC1101

Execution oracle
concrete.rs · code-disjoint
per-path concrete replay
witnesses soundness